

Exam Date & Time: 22-Nov-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VII Semester - End Semester Examination - Nov / Dec -2024

Advanced Cryptography [CSE-4059]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1)

Prove that $Z_2 / (x^2 + x + 1)$ is isomorphic to the finite field F_4 . (5)

 - A)
 - B)
 - a) Show that the set $\{5, 15, 25, 35\}$ is a group under multiplication modulo 40.
 - b) What is the identity element of this group?
 - c) Can you see any relationship between this group and $U(8)$ the group consisting of invertible elements in Z_8 ? (5)
- 2)

Let $f(x) = x^3 + 2x^2 + 1$ and $g(x) = 2x^3 + x + 2$ over F_3 (Finite field with 3 elements). Perform the polynomial addition $f(x) + g(x)$. (5)

 - A)
 - B)

Use Fermat's Little Theorem to find $3^{100} \bmod 7$. (5)
- 3)

Consider a modified ElGamal cryptosystem over F_{47} , with the following parameters:

 - A)

Prime $p=47$, generator $g=5$. Alice's private key $a=15$, public key $A = g^a \bmod P = p^a \bmod 47$. Bob wants to send a message $m=22$ to Alice. Bob chooses a random integer $k=9$. (5)

 - a) Compute the ciphertext pair (c_1, c_2) for Bob's encryption.
 - b) Show how Alice decrypts the ciphertext and recovers the message.
 - B)

In the ElGamal cryptosystem, given $p=23$, generator $g=5$, public key $y=8$, ciphertext pair $(c_1, c_2)=(10, 4)$, and private key $x=6$, decrypt the ciphertext to find the plaintext message. (5)
- 4)

How does the arithmetic of elliptic curves differ over the finite field $GF(p)$ compared to (5)

real numbers? Give an example of a point addition operation over $GF(23)$ for an elliptic curve.

- A)
 - B) Given a 128-bit block cipher and the key $K=0x2B7E151628AED2A6ABF7158809CF4F3C$, compute the CMAC for the message $m=Hello$. Assume padding is required to make the message length a multiple of the block size (128 bits). Show the steps involved in computing the MAC. (5)
- 5) Consider a one-time password system where the OTP is generated using a shared secret key $K=0xABCD1234$ and a counter value of $C=5$. If the OTP is generated using HMAC-SHA256, calculate the OTP for the counter value $C=5$. Show the intermediate steps, including the result of the HMAC-SHA256 function. (5)
- A)
 - B) **Scenario:** A new online banking application has been developed that uses both data-origin and entity authentication mechanisms to enhance security for its users. The application employs one-time passwords (OTPs) and a challenge-response authentication system to verify users.
 - i) Describe how the online banking application utilizes data-origin authentication and entity authentication. What are the benefits of using both methods in this scenario? (5)
 - ii) Suppose Alice wants to log in to her online banking account. She receives a challenge in the form of a nonce from the server. Explain how the challenge-response protocol works in this situation, detailing the steps Alice must take to authenticate herself securely.

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